Probe-Then-Act: Learning Material-Adaptive Manipulation through Active Tactile Exploration in Multi-Physics Simulation

Anonymous Authors

***Abstract*—The action that reveals hidden dynamics can also change the state a robot must control. We study this information–intervention tradeoff in contact-rich material manipulation using PROBE-THEN-ACT, a deliberately minimal experimental probe: a fixed interaction elicits a material response, an encoder summarizes the proprioceptive and simulator-derived aggregate-particle trace, and a residual policy acts from the perturbed state. On a Genesis edge-push benchmark spanning sand, snow, and elastoplastic media, all learned policies train on sand and transfer zero-shot. The central observation is an asymmetric sign pattern. On OOD elastoplastic, PROBE-THEN-ACT improves transfer efficiency by 14.7 percentage points (60.7% vs. 46.0%), reduces spillage by 14.7 points (39.3% vs. 54.0%), and raises success by 16.7 points over reactive PPO. Matched ablations show that removing the probe falls below the reactive baseline, while removing only the encoder output returns to roughly baseline transfer and remains below full PROBE-THEN-ACT. On ID sand, two sand parameter shifts, and snow, the same probe-conditioned policy underperforms by 13–22 points. We organize this pattern through a falsifiable recoverability hypothesis: active context acquisition is net useful when the probe-induced state shift relaxes on the task timescale, and costly when the intervention leaves persistent disruption. The claim is scoped to this diagnostic benchmark and motivates recoverability as a testable organizing variable for future material families and probe designs.**

***Index Terms*—robot learning, active perception, material-adaptive manipulation, multi-physics simulation, deformable object manipulation**

I. INTRODUCTION

ROBOTIC tool use on deformable and flowable media—scooping granular piles, pushing viscous pastes, transferring cohesive snow-like aggregates—requires adapting manipulation to hidden physical properties that are not visible in a single snapshot. Two materials that look nearly identical to a camera can differ by an order of magnitude in stiffness, cohesion, or flow resistance, and the manipulation strategy that succeeds on one of them can fail catastrophically on the other [1]–[3]. A robot that relies only on instantaneous observations therefore faces a dynamics-ambiguity problem: the same observed state can correspond to many distinct transition models.

A common response is to train reactive policies with domain randomization [4], [5] or to use a history-conditioned recurrent policy [6], [7]. These approaches trade breadth for depth—they spread the policy thin across many material instances without ever explicitly inferring what material the robot is currently manipulating. An alternative is discrete material classification followed by per-class policy routing [8], which requires a closed set of material labels and brittle perception. Neither strategy offers an explicit mechanism for the robot to *ask the physics a question* before committing to a manipulation plan.

We take a different position: the information-gathering interaction is itself a physical intervention on the task state. Humans probe unfamiliar substances before picking them up and use the response to pick a grasp and a motion profile, but the probe can also compact, displace, or otherwise alter the material. We operationalize this intuition in PROBE-THEN-ACT (Figure 1). PROBE-THEN-ACT is deliberately minimal: the probe is a fixed open-loop trajectory, the encoder is a small MLP, and the residual policy is standard PPO. The active component is the budgeted allocation of an initial interaction segment and the learned downstream use of its response; the probing action itself is not optimized as an information-gathering policy. The contribution of this paper is therefore empirical and diagnostic—a study of the sign pattern created when information gain and probe-induced state perturbation compete—not a new inference architecture.

1) **Probe phase.** The robot executes a short, fixed open-loop sequence of exploratory contacts with the material while the task residual is held at zero. The resulting proprioceptive and aggregate-particle trace implicitly encodes the material's response to applied forces.
2) **Encoder.** An MLP with mean-pooling maps the probe trace to a latent task-conditioning vector $z \in \mathbb{R}^{16}$ and a learned scale σ. We refer to (z, σ) as a "belief" for brevity, but the pair is not a calibrated posterior: there is no KL regularizer, prior, or auxiliary classification loss; z is whatever representation of the probe response is task-useful under the end-to-end PPO objective.
3) **Task policy.** A joint-residual PPO policy is conditioned on (s_t, z, σ) and emits joint-space residuals on top of a scripted edge-push base trajectory. The base trajectory sidesteps differentiable-IK coupling artifacts we observed in the simulator [9].

a) The recoverable-deformation hypothesis.: Probing is not free: every probe step displaces particles, breaks contacts, and consumes task time. We offer a physics-grounded interpretation of when this cost is likely to pay off, treated throughout as an *interpretive hypothesis* rather than a demonstrated principle:

> *Active probing should be expected to help downstream manipulation only when the material's response to a probe is* ***recoverable****—the probed configuration relaxes back, on the task timescale, to a state*

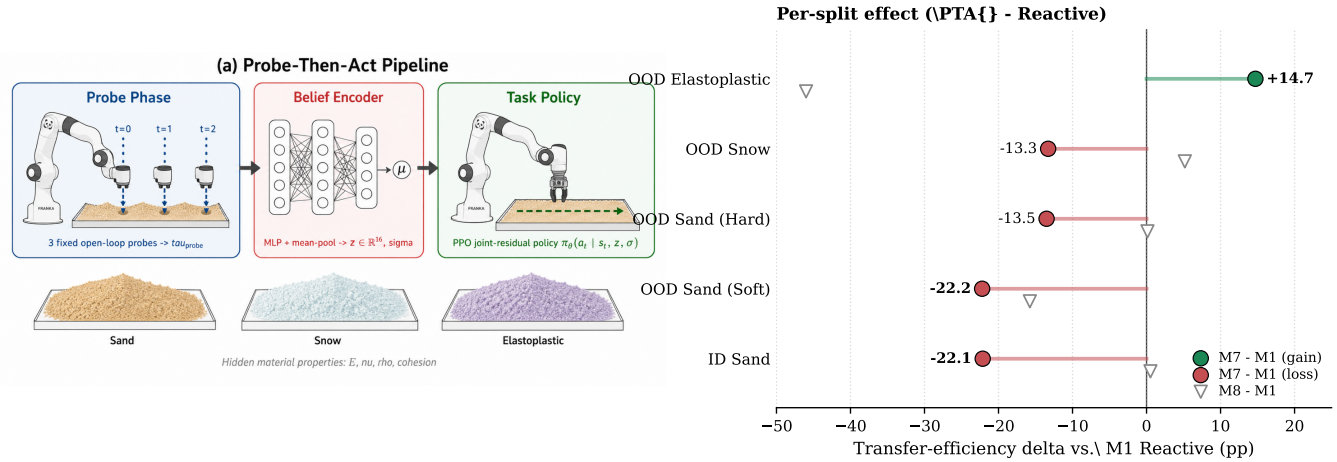

Fig. 1: **PROBE-THEN-ACT framework and per-split effect.** Left: schematic illustration of the two-stage architecture — (i) the robot executes a short fixed open-loop probing phase, (ii) an encoder maps the probe trace to a latent task-conditioning vector z with a learned scale parameter σ (not a calibrated posterior), (iii) the task policy is conditioned on (s_t, z, σ) to produce adaptive joint-space residual actions. Material piles below the schematic illustrate the three Genesis MPM families used in the benchmark; the schematic is for explanation and is not a Genesis render (Genesis renders appear in Figure 2). Right: per-split transfer-efficiency delta of PROBE-THEN-ACT (M7) versus the matched reactive baseline (M1), in percentage points; hollow downward markers show the single-seed M8 privileged-parameter diagnostic delta. PROBE-THEN-ACT is positive on OOD elastoplastic ($+14.7$ pp) and negative on ID sand, snow, Sand-Hard, and Sand-Soft (-13 to -22 pp), while the M8 diagnostic is -46 pp on elastoplastic. M1/M7 are means over 3 training seeds with 10 evaluation episodes per seed; this overview shows mean deltas only and the comparison is descriptive. Matched-seed dispersion is shown in Figure 5; absolute values for all three methods are in Figure 3.

the policy can still exploit—and to be net-negative when probe-induced perturbations are ***irreversible****.*

Elastoplastic media in our benchmark exhibit recoverable elastic rebound; granular sand (especially under low cohesion) exhibits irreversible rearrangement. Snow is cohesive in appearance but remains a negative case under contact loading, consistent with a persistent compacted or displaced region rather than a relaxed rebound. The hypothesis therefore anticipates, and our experiments are consistent with, gains on elastoplastic and losses on the persistent-perturbation splits.

b) Empirical evaluation.: We evaluate PROBE-THEN-ACT on a reproducible cross-material edge-push benchmark in Genesis [9] spanning sand (granular, 32% scripted transfer), snow (cohesive, 87%), and elastoplastic medium with recoverable elastic rebound (70%), plus two parameter-shifted sand variants. The 55-percentage-point scripted-transfer spread is discriminative by design. All methods are trained on the hardest split (sand) only and evaluated zero-shot on the other four, so the encoder is asked to *extrapolate* rather than to interpolate over a labeled material mix.

On the elastoplastic split, PROBE-THEN-ACT achieves 60.7% transfer versus 46.0% for a matched reactive PPO baseline (M1)—14.7pp absolute, 32% relative—accompanied by a 14.7pp reduction in spillage (39.3% vs. 54.0%; Table I, Figure 3). Matched ablations clarify the source of this improvement: removing the probe drops transfer by 27.6pp and falls below M1, while removing only the encoder output drops 14.4pp from full PROBE-THEN-ACT and returns to roughly M1-level transfer (Table II). On ID sand, snow, and two sand parameter shifts, PROBE-THEN-ACT underperforms by 13–22pp, including a 22pp loss on in-distribution sand. We treat this as an observed sign pattern: active probing is positive on the recoverable split and negative where the probe-induced perturbation persists. A complementary observation: a single-seed privileged-parameter diagnostic (M8) that receives ground-truth material parameters scores 0% transfer on elastoplastic while remaining strong elsewhere. In our sand-only-trained, residual-control setup, a passive parameter vector did not unlock a learned response curve for elastoplastic rebound while a probe-induced response did.

c) Contributions.: We make three contributions, framed as empirical and diagnostic:

1) A **diagnostic sign pattern**: on a discriminative cross-material benchmark, a minimal probe-then-act policy improves the recoverable elastoplastic split (+14.7pp transfer, −14.7pp spill, +16.7pp success) while underperforming by 13–22pp on ID sand, snow, and two sand parameter shifts; matched ablations show that skipping the probe falls below the reactive baseline, while dropping only the encoder returns to roughly baseline transfer and remains below full PROBE-THEN-ACT.
2) A **recoverable-deformation hypothesis**, offered as a physically motivated candidate boundary condition for active probing and a testable prediction for future work on additional viscoelastic and compacting granular media.
3) A **reproducible cross-material benchmark** in Genesis with five splits (55pp of scripted-transfer spread), standardized 3-seed protocol, per-seed metrics, and ablation infrastructure; code and seeds will be released upon acceptance.

d) Scope of claims.: PROBE-THEN-ACT as evaluated here uses a fixed open-loop probe, observations that include simulator-derived aggregate-particle statistics, and an end-to-end task-conditioning latent without calibration or material-label supervision; all training is on sand. This scopes the paper as a *diagnostic* study of when a minimal probe-conditioned policy helps and as a benchmark for future sensing and robustness work.

The rest of the paper is organized as follows. Section II positions our work against active perception, deformable manipulation, system identification, and multi-physics benchmarks; Section III formalizes PROBE-THEN-ACT; Section IV describes the benchmark and evaluation; Section V reports the main results and ablations; Sections VI and VII discuss scope, limitations, and conclude.

II. RELATED WORK

Our work sits at the intersection of active perception, deformable-object manipulation, online system identification, and cross-material simulation benchmarks. The architectural pieces of PROBE-THEN-ACT—a fixed open-loop probing segment, an MLP encoder of the probing trace, and a context-conditioned residual policy—are deliberately minimal and individually well precedented. We position PROBE-THEN-ACT not as a new inference architecture but as a minimal instantiation of active in-episode material identification used to isolate when probing helps or hurts deformable-material manipulation. Below we review each adjacent line of work, emphasizing what PROBE-THEN-ACT *infers*, what it *conditions on*, and what it is *evaluated against*, rather than claiming algorithmic novelty over each.

a) Active perception for manipulation.: Robots have long been studied as embodied inference agents that gather information by acting on the world. Early work on active vision [10] and exploratory procedures for haptic object recognition [11] established the principle that specific motions yield specific physical information. In modern learning-based manipulation, tactile exploration has been used to estimate local object geometry and contact properties [8], [12], to inform regrasping policies [13], and to estimate object pose through pushing [14], [15]. Recent interactive perception systems actively choose visuo-tactile pushing or pulling actions to infer physical object properties [16], and tactile-informed dynamics models use contact observations for dense non-prehensile manipulation [17]. These works motivate probing as a physical measurement operation, but many focus on rigid objects, local contact properties, or closed-loop controllers where the measurement action's downstream damage is not isolated. PROBE-THEN-ACT extends this line by targeting continuous, bulk physical properties of *deformable* media (stiffness, cohesion, plasticity) while treating the probe itself as an intervention whose disturbance may or may not relax before the downstream push.

b) Deformable and granular manipulation.: Manipulating soft and flowable materials has become a focus area as differentiable and particle-based physics engines have matured. Environments such as PlasticineLab [1] and DiffSkill [2] study skill learning on plasticine-like media via differentiable MPM simulation, and RoboCook-style systems [3], [18] study tool use on dough-like materials. Recent work also adapts graph-based dynamics to material properties [19], identifies elastoplastic parameters from robot interaction [20], and scales granular robotic-task simulation [21]. A common feature of these works is that training and evaluation use the same material family, learn or calibrate an explicit dynamics model, or treat material variation via blanket domain randomization. Our benchmark instead organizes evaluation around a *cross-material* split structure—training on the hardest granular material and evaluating zero-shot on cohesive and elastoplastic media—so that the role and cost of material-specific probing can be isolated.

c) Online system identification and adaptation.: Adapting policies to unseen dynamics has been approached through meta-learning of dynamics and policies [22], [23], context-conditioned policies that infer a latent task embedding from recent history [24], [25], and rapid online model adaptation [26], [27]. Robotic sim-to-real pipelines have combined domain randomization with online adaptation of a low-dimensional parameter vector [5], [28]. Active system-identification methods go further by optimizing exploratory behavior to collect data for simulator or parameter refinement [29]. These approaches typically assume that the information needed to adapt is present in the course of normal task execution, or that the identification signal's physical cost is secondary to the parameter estimate. PROBE-THEN-ACT instead *budgets* a dedicated, short probing phase at the start of each episode whose only purpose is to elicit material response. The conditioning latent and the residual policy are otherwise standard. The contribution we attach to this design choice is empirical, not architectural: we use it as a controlled lens to ask when separating "what material am I holding?" from "how do I act?" helps versus hurts, and our results show that the answer depends on whether probe-induced perturbations relax on the task timescale.

d) Multi-physics simulation and cross-material benchmarks.: Recent simulators have made multi-material robotics research practical. Genesis [9] provides a unified framework covering rigid body, MPM, SPH, and FEM physics. Isaac Gym [30] and Brax [31] are widely used for GPU-accelerated rigid-body RL but do not natively support deformable media. Benchmarks such as SoftGym [32], PlasticineLab [1], and GranularGym [21] evaluate policies on deformable or granular tasks but typically fix the material family per task. Our benchmark complements these by pairing a single task definition with a deliberately discriminative material sweep (55pp of baseline variation across families), along with parameter-shifted in-family OOD splits and a standardized evaluator. The goal is a narrow, reproducible testbed in which the specific contribution of material-adaptive components—as opposed to task-specific engineering—can be read off the numbers.

III. METHOD

We formalize PROBE-THEN-ACT as a two-stage decision process that augments a standard finite-horizon MDP with an explicit probing phase and a latent task-conditioning step.

The formulation makes explicit that probing changes both the information available to the network and the physical state distribution faced by the task policy.

A. Problem Formulation

An episode consists of a robot interacting with a material instance $m \in \mathcal{M}$ drawn from a distribution over hidden physical properties (Young's modulus E, Poisson ratio ν, density ρ, cohesion, and plasticity parameters). The robot receives observation $s_t \in \mathbb{R}^{d_s}$ at step t and applies action $a_t \in \mathbb{R}^{d_a}$. The observation contains proprioceptive state (joint positions, end-effector pose, time fraction) and aggregate particle statistics computed from simulator state—mean y-coordinate, transfer fraction into the target region, and spill fraction off the platform—but *not* the ground-truth material parameters m. The task reward combines transfer, spill, and success terms defined for our edge-push benchmark; Genesis provides the simulator.

Our goal is to learn a policy $\pi_\theta(a_t \mid s_t, z, \sigma)$ that maximizes expected episode return over the material distribution, where (z, σ) is a latent task-conditioning vector produced by an encoder q_ϕ from a short probing rollout. We refer to (z, σ) as a "belief" for brevity but do not interpret it as a calibrated posterior: σ is a learned scale used at training time, not a posterior standard deviation. m is never observed; z must be inferred entirely from the probing interaction, and its usefulness depends on the post-probe state as well as the encoded material response.

B. Two-Stage Structure

An episode of horizon H is split into a probing phase of length $H_p = 3$ and a task phase of length $H - H_p$.

a) Probe phase.: During the first H_p steps the task residual is held at zero and the robot executes a fixed, open-loop probing trajectory a_t^{probe} over a scripted approach. This yields a probing trace

$$\tau_{\text{probe}} = \{(s_0, a_0^{\text{probe}}), \ldots, (s_{H_p-1}, a_{H_p-1}^{\text{probe}})\}, \tag{1}$$

which is a $H_p \times d_{\text{probe}}$ tensor whose rows capture how the material responded to each applied action. We deliberately use a fixed probe schedule rather than a learned probe policy: it isolates the contribution of the *encoder* and keeps the probe-vs-task credit assignment clean.

b) Encoder.: A two-layer MLP with hidden size 256 maps each probe step independently to a feature vector, mean-pools across steps, and emits $(\mu_z, \log \sigma_z^2) \in \mathbb{R}^{16} \times \mathbb{R}^{16}$ via a linear head. We sample $z \sim \mathcal{N}(\mu_z, \text{diag}(\sigma_z^2))$ at training and use $z = \mu_z$ at evaluation. The encoder is trained jointly with the policy via the PPO loss only—no classification objective, KL regularizer, or calibration loss on σ_z.

c) Task policy.: The task policy π_θ is a feedforward Gaussian policy over joint-space residuals, with state $[s_t, \mu_z, \sigma_z]$. It emits a 7-DoF residual Δq_t that is added to a scripted edge-push base trajectory q_t^{base}:

$$q_t = q_t^{\text{base}} + \eta \cdot \Delta q_t, \tag{2}$$

where $\eta = 0.05$ is a residual scale that empirically balances exploration with base-trajectory preservation. Applied joint targets are commanded through `set_qpos`, bypassing inverse kinematics coupling artifacts we observed in Genesis at the home configuration.

C. Training Objective

Both the encoder q_ϕ and the policy π_θ are trained end-to-end with PPO [33]. The probe phase runs first; the encoder produces (μ_z, σ_z) from τ_{probe}; the task phase is rolled out with $\pi_\theta(\cdot \mid s_t, \mu_z, \sigma_z)$. Advantages and returns are computed over the task phase only, so z is shaped indirectly by its downstream effect on returns. Because the probe trajectory is fixed open-loop, gradients do not flow into the probing actions; the encoder learns to extract task-relevant signal from a pre-specified probing response rather than to optimize the probe.

D. Ablation Variants

We study two ablations that remove one component at a time while keeping everything else fixed:

- **No-Probe.** The probing phase is skipped; the encoder receives a zero trace and produces a degenerate z. This isolates the contribution of actually executing probing actions.
- **No-Belief.** The probe phase still runs, but the encoder output is dropped and the task policy conditions only on s_t (as in the reactive baseline, but trained with the same pipeline). This isolates the contribution of explicitly encoding the probing information.

Together these ablations test whether the gain stems from probing itself, from belief conditioning, or from incidental factors in the training pipeline.

IV. Experimental Setup

A. Benchmark Design

We build the cross-material benchmark in the Genesis multi-physics simulator [9] around a single *edge-push* task. A Franka Panda arm (7 DoF) equipped with a flat scoop end-effector must push a pile of particles from an elevated source platform across a raised edge into a target bin, with spill onto the floor penalized. This task is deliberately simple kinematically—it is essentially a horizontal push followed by a settle—so that differences across methods can be attributed to material-adaptive behavior rather than to task-specific engineering.

Table III summarizes the five material splits. Figure 2 shows the corresponding Genesis renders; the same Franka, scoop, source platform, target bin, and pile placement (Config D, 5.5 cm from the open edge) are used for every split, so cross-split differences come from material parameters alone. We use Genesis MPM at grid density 128 with 25 substeps per control step (control $\Delta t = 2 \times 10^{-3}$ s).

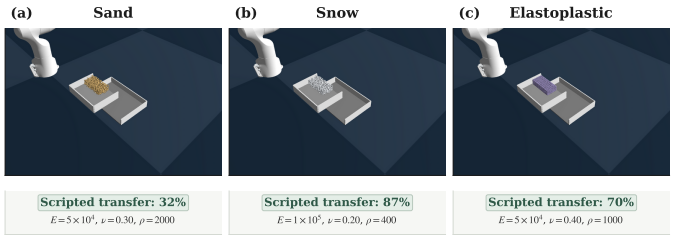

Fig. 2: **Genesis edge-push benchmark across material families.** Rendered frames from the Genesis MPM simulator after the pile has settled on the source platform. Scene geometry (Franka Panda, source platform, target bin, particle pile placement following Config D, 5.5 cm from the open edge) is identical across panels; only the MPM material parameters differ (overlaid bottom-right of each panel: Young's modulus E in Pa, Poisson ratio ν, density ρ in $\mathrm{kg\,m^{-3}}$). Equilibrium shapes are visibly distinct—granular sand spreads, cohesive snow holds together loosely, and the elastoplastic medium retains a tight rebounded block. Bottom-left (green) tag reports the scripted-controller transfer efficiency for that material as a discriminative-difficulty reference. Particle surface colours are chosen per panel for visual contrast and have no effect on dynamics.

a) Material choices.: The three material families span clean and boundary regimes. *Sand* (granular, $E = 5 \times 10^4$, $\nu = 0.3$, $\rho = 2000$) is the training material and the hardest—a scripted baseline achieves only 32% transfer because particles deform under the scoop and flow backward. *Elastoplastic* ($E = 5 \times 10^4$, $\nu = 0.4$, $\rho = 1000$) is the clean recoverable case: the pile partially rebounds after each push, so a fixed residual over-corrects. *Snow* (cohesive, $E = 1 \times 10^5$, $\nu = 0.2$, $\rho = 400$) is deliberately a boundary case. Cohesion makes the scripted push easy (87% transfer), but contact loading can leave a compacted or displaced region rather than a rebounded one, so recoverability is not equated with cohesion or task ease. We use "elastoplastic" throughout to refer to this Genesis material model and reserve "viscoelastic" as a description of the rebound behavior it exhibits, rather than as an independent material class. The 55pp spread between hardest and easiest is *discriminative* by design, forcing methods to actually adapt rather than memorize a single push profile.

b) OOD parameter splits.: In addition to family-level OOD (snow, elastoplastic), we evaluate on parameter-shifted sand variants: **Sand-Hard** ($E = 8 \times 10^4$, 60% stiffer) and **Sand-Soft** ($E = 2 \times 10^4$, 60% softer). These probe within-family generalization and serve as controls: if a method's gain on elastoplastic is driven by generic "any-OOD" robustness, the parameter splits should show similar gains.

B. Evaluation Protocol

All methods train on ID sand only and are evaluated zero-shot on the four other splits, testing *extrapolation* rather than interpolation over a labeled material mix. Each (method, seed, split) is scored by 10 deterministic-policy episodes. Metrics: (i) transfer efficiency (fraction of particles delivered to the target), (ii) spill ratio (fraction off the platform/outside the target), and (iii) success rate (fraction of episodes exceeding 30% transfer; the 30% threshold is a level the scripted controller exceeds on snow and elastoplastic but not on sand, distinguishing bulk delivery from stirring). M1, M7, and the M7 ablations use three seeds (42, 0, 1) and report mean $\pm$ std; M8 is scored as a single-seed diagnostic baseline. NaN rollouts (rare, from Genesis contact blowups) count as zero-transfer/unit-spill failures so a method cannot benefit from crashing the simulator.

C. Compared Systems

We compare five systems, all trained with identical 500 k-step PPO budgets and matched hyperparameters (residual scale 0.05, observation: proprio + particle statistics):

- **M1: Reactive PPO**—standard reactive baseline conditioned on s_t only. No probe, no belief.
- **M7: PROBE-THEN-ACT (ours)**—the full two-stage architecture of Section III.
- **M8: Privileged-Parameter Baseline**—a sand-only-trained residual policy that additionally receives ground-truth material parameters (7-D privileged observation) concatenated to s_t, but is otherwise matched to M1/M7. It tests parameter access under the same sand-only training protocol.
- **M7-NoProbe**—ablation that skips the probing phase; encoder sees a zero trace.
- **M7-NoBelief**—ablation that runs the probe but drops the encoder output.

M1 and M7 are the core comparison. M8 is a privileged *controlled* baseline—matched architecture and training protocol, plus ground-truth parameters—so it isolates the value of those parameters under sand-only training rather than serving as a universal informational upper bound. The two ablations localize PROBE-THEN-ACT's gain across its components.

D. Implementation Details

All runs use PPO from Stable-Baselines3 [34] with MLP policy/value heads of size $[256, 256]$, lr 3×10^{-4}, batch 64, 10 epochs/update, $\gamma = 0.99$. The 7-DoF residual is added to a 410-step scripted edge-push trajectory followed by an 80-step settle; joint targets are commanded via `set_qpos`, which in our environment bypasses single-step DLS-IK coupling artifacts. Code, configurations, and seeds will be released upon acceptance.

V. RESULTS

We organize the results around the information–intervention tradeoff: where does the probe's information gain dominate the physical state shift, where does the state shift dominate, and does the resulting sign pattern align with the recoverable-deformation hypothesis from Section I?

A. Main Cross-Material Results

Table I and Figure 3 report transfer efficiency across all five material splits. M1 and M7 are averaged over three random seeds; M8 is a single-seed diagnostic row. We observe a clear, asymmetric pattern.

TABLE I: Cross-material transfer efficiency (%) across five evaluation splits. For M1 and M7, entries report mean ± seed standard deviation over 3 training seeds (10 deterministic episodes per seed). M7 leads on OOD elastoplastic by 14.7 pp; per-seed dispersion on elastoplastic is ~ 40 pp and is shown explicitly in Figure 5. M8 is a single-seed diagnostic privileged-parameter baseline; no seed standard deviation is reported.

Method	ID Sand	OOD EP	OOD Snow	OOD Hard	OOD Soft	Avg OOD
M1: Reactive	64.1±8.0	46.0±40.0	55.9±22.8	62.3±7.4	68.7±10.7	58.2
M7: Probe-Then-Act (Ours)	41.9±9.1	**60.7±43.3**	42.6±18.4	48.8±9.9	46.5±5.9	49.7
M8: Privileged-Param (single seed)	64.6	0.0	61.1	62.4	53.0	44.1

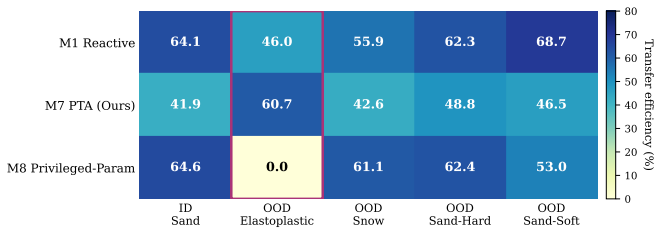

Fig. 3: **Absolute transfer-efficiency landscape.** 3-method × 5-split heatmap of mean transfer efficiency (%). M1 and M7 cells are 3-seed means (10 deterministic episodes per seed; per-seed dispersion in Figure 5); M8 is scored from a single trained seed and serves as a diagnostic row. The OOD-elastoplastic column is outlined to flag the conditional finding: PROBE-THEN-ACT (M7) attains 60.7% transfer there versus 46.0% for the reactive baseline (M1) and 0% for the privileged-parameter baseline (M8). The complementary effect plot in Figure 1 reports the same data as deltas relative to M1. M8 here is a sand-only-trained residual policy that additionally receives ground-truth $(E, \nu, \rho, \dots)$; its elastoplastic collapse contains zero NaN events. Across all (method, split, seed) combinations reported in Tables I and II, total NaN-rollout count was zero.

a) Gain on the elastoplastic (recoverable-rebound) split.: On the OOD elastoplastic split, PROBE-THEN-ACT (M7) achieves 60.7% transfer (3-seed mean), compared to 46.0% for the matched reactive baseline M1—an absolute gain of 14.7 percentage points, or 32% relative. Spill drops by exactly 14.7 percentage points (39.3% vs. 54.0%), and success rate (fraction of episodes exceeding 30% transfer) rises from 50.0% to 66.7%. The improvement is therefore consistent across all three metrics. Given the high per-seed variance, we discuss the statistical strength of this gain in Section V-C before drawing conclusions.

b) Opposite sign on persistent-perturbation splits.: On ID sand, M1 outperforms PROBE-THEN-ACT by 22.2pp (64.1% vs 41.9%); on parameter-shifted sand variants, M1 leads by 13–22pp; and on cohesive snow, M1 leads by 13.3pp (55.9% vs 42.6%). These negative signs are part of the empirical finding: the same probe that helps on recoverable elastoplastic dynamics is costly on splits where the contact perturbation persists. This is the central asymmetry that the recoverable-deformation hypothesis is designed to explain (Section V-D).

c) Privileged-parameter collapse on the elastoplastic split.: The privileged-parameter baseline M8 receives ground-truth material parameters and is otherwise matched to M7. Unlike M1 and M7, M8 was scored from a single trained seed in this evaluation; we treat its row in Table I and Figure 3 as diagnostic rather than as a 3-seed comparison. M8 is near M1 on ID sand (64.6%), snow (61.1%), and the parameter-shifted sand variants (62.4%, 53.0%). On OOD elastoplastic, its single scored seed collapses to exactly 0% transfer. Diagnostic checks rule out simulator crashes (zero NaN events; M1 and M7 produce nonzero transfer on the same env state) and are consistent with residual over-correction of the elastoplastic rebound by a head shaped on sand, where rebound is negligible. In our matched setup, a policy with passive access to $(E, \nu, \rho, \dots)$ could not extrapolate from sand to elastoplastic, while PROBE-THEN-ACT's in-episode probing trace did.

TABLE II: Ablation study on OOD Elastoplastic split. Removing the probe phase drops mean transfer by 27.6pp and falls below M1; removing the belief encoder drops 14.4pp from full PROBE-THEN-ACT and returns to roughly M1-level transfer. Success is the aggregate fraction over 30 episodes; equal transfer/spill standard deviations occur when per-seed spill is the complement of transfer.

Variant	Transfer (%)	Spill (%)	Success (%)
M7 Full (Ours)	**60.7±43.3**	39.3±43.3	66.7
M7 No-Probe	33.1±46.8	51.7±40.6	33.3
M7 No-Belief	46.3±40.1	53.7±40.1	46.7
M1 Baseline	46.0±40.0	54.0±40.0	50.0

B. Ablation Study

Table II and Figure 4 break down PROBE-THEN-ACT's elastoplastic gain by component, holding all other training pipeline elements fixed.

a) Probe execution carries signal; conditioning closes the gap.: Removing the probing actions (M7-NoProbe) collapses transfer to 33.1%—a 27.6pp drop from M7 Full and 12.9pp *below* the reactive baseline. Removing only the belief encoding while keeping the probing phase (M7-NoBelief) preserves 46.3% transfer, matching the reactive baseline but 14.4pp below PROBE-THEN-ACT Full. Spill follows the same ordering, and success rate exhibits an even sharper separation: M7 Full succeeds in 66.7% of elastoplastic episodes, versus 33.3% for No-Probe and 46.7% for No-Belief.

This pattern argues against a pure wrapper-stack or model-capacity explanation for the full elastoplastic gain: No-Probe is worse than M1, and No-Belief remains near M1 rather than approaching full PROBE-THEN-ACT. The result instead implies that the probe-induced state shift is harmful unless the downstream policy can exploit a useful conditioning signal.

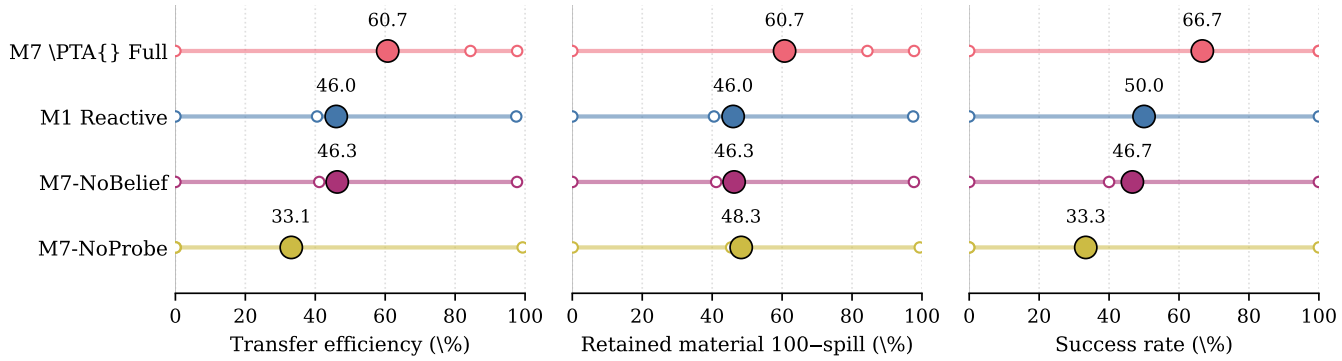

Fig. 4: **Ablation study on OOD elastoplastic.** Cleveland dot-plot matrix; each row is a method, each column a metric. Filled markers are 3-seed means; the horizontal whisker spans the per-seed minimum–maximum, and the small white dots show individual seed values (10 deterministic episodes per seed). Spill is plotted as "Retained material" (100 − spill) so that all three columns are higher-is-better. No-Probe falls below the reactive baseline on transfer; No-Belief is near the reactive baseline on transfer but remains below full PROBE-THEN-ACT.

b) In this architecture, removing the probe is more damaging than removing the encoder output.: On the elastoplastic split, removing the probe drops transfer by 27.6pp while removing only the encoder output drops it by 14.4pp. Within PROBE-THEN-ACT's architecture and on this split, this is consistent with the physical probing interaction—which temporarily reshapes the particle pile and elicits an elastoplastic rebound—supplying signal that is not recoverable from passive observation alone. We do not generalize this ordering beyond the present architecture, ablation pair, and split: a different probe schedule, encoder, or material could in principle invert it.

C. Statistical Reading of the Elastoplastic Gain

With $n = 3$ seeds, 10 deterministic episodes per (method, seed, split), and a per-seed transfer standard deviation of ∼40pp on elastoplastic, we report effect sizes and per-seed metrics rather than collapse the result into a significance claim. Figure 5 shows that PROBE-THEN-ACT's seed-mean exceeds M1's on two of the three matched seed pairs (the third is closer). The mean gain is supported by triangulation—higher mean transfer, lower mean spill (also 14.7pp), higher success rate—and by ablations where skipping the probe falls below M1 while dropping only the encoder returns to roughly M1-level transfer. We release per-seed numbers and the benchmark code to make additional seeds straightforward to add.

D. Alternative Explanations and Recoverability

Figure 3 shows an asymmetric pattern: PROBE-THEN-ACT helps on elastoplastic and hurts elsewhere. Two simpler accounts must be considered. First, if the elastoplastic gain were only an OOD-distance or capacity effect, then the same wrapper should also help on the sand parameter shifts; instead Sand-Hard and Sand-Soft remain negative, No-Belief stays near M1 on elastoplastic, and No-Probe drops below M1. This weakens but does not eliminate capacity/OOD-distance explanations because our material distances are not metric-matched. Second, sand-only training may itself make probe perturbations harmful on sand-like regimes; the ID-sand loss is consistent with this and is part of the intervention-cost signal. Seed noise remains the third alternative: elastoplastic has ∼40pp dispersion. We therefore treat recoverability as the current organizing hypothesis, to be distinguished by matched material-distance sweeps, equal-capacity dummy probes, non-sand training sources, and direct post-probe recovery measurements.

a) Elastoplastic (recoverable rebound): gain.: A probe displaces particles, but the displacement partially relaxes during the brief settle before the task phase, so rebound information survives in the trace while substrate disruption is largely undone. This is the regime in which the hypothesis predicts a favorable probe trade, and PROBE-THEN-ACT achieves the 14.7pp transfer gain.

b) Granular (ID sand and sand variants): loss.: Granular media without cohesion rearrange irreversibly: probes push grains forward and shift the centroid in a way that does not relax. The 22.2pp loss on ID sand is the cleanest signal—the policy was trained on this exact material yet loses because probing has degraded the substrate. Sand-Hard and Sand-Soft show the same direction at similar magnitude, weakening a generic "any-OOD" explanation though not isolating recoverability from intra-family confounders.

c) Cohesive snow: persistent-perturbation loss.: Cohesive snow holds together against probe perturbations but does not actively rebound before the downstream push, leaving a compacted or displaced region in this benchmark. PROBE-THEN-ACT's 13.3pp snow loss is therefore the boundary case anticipated in Section IV: it refines the hypothesis toward post-perturbation recoverability rather than material labels such as "cohesive" versus "granular."

Figure 5 plots the per-seed numbers behind the seed-mean comparison.

VI. DISCUSSION

A. The Recoverable-Deformation Hypothesis as a Testable Heuristic

The present benchmark turns recoverability into a testable heuristic rather than a settled law: the positive evidence

TABLE III: Cross-material benchmark composition. Scripted baseline transfer varies by 55pp, confirming discriminative difficulty. E: Young's modulus; ν: Poisson ratio.

Split	Material (E, ν)	Scripted	Role
ID	Sand (5×10^4, 0.3)	32%	Training
OOD-Mat.	Snow (1×10^5, 0.2)	87%	Unseen family
OOD-Mat.	Elastoplastic (5×10^4, 0.4)	70%	Unseen family
OOD-Par.	Sand-Hard (8×10^4)	–	Param. shift
OOD-Par.	Sand-Soft (2×10^4)	–	Param. shift

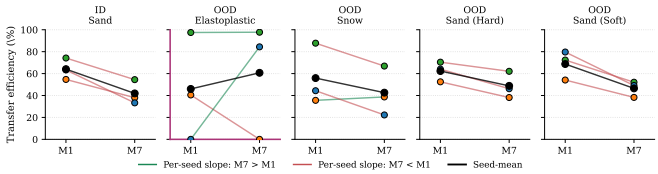

Fig. 5: **Paired-seed slope chart, M1 vs. PROBE-THEN-ACT.** One panel per material split; each thin line connects matched-seed transfer efficiencies (M1 left, PROBE-THEN-ACT right) for the same training seed. Green slopes indicate PROBE-THEN-ACT $>$ M1 on that seed, red slopes indicate the reverse; the heavy black line connects 3-seed means. The OOD-elastoplastic panel (highlighted) is the only split where the seed mean rises and where PROBE-THEN-ACT exceeds M1 on 2 of 3 matched seed pairs (the third is closer). On the four other splits all per-seed slopes go down. Per-seed dispersion is largest on elastoplastic (~ 40 pp), so we report seed-level values directly in addition to means (Section V-C).

comes from a single elastoplastic family, and recoverability is not independently varied from task ease, particle density, control stability, sand-only training, or the simulator's specific elastoplastic constitutive model. We offer it as a physically grounded interpretation of the asymmetry and as a design rule for targeted follow-up experiments.

If the hypothesis holds more broadly, the implied prescription is to allocate probe time to a material if and only if the probe-induced perturbation relaxes back on the timescale of the subsequent task. Each row of the candidate prescription below beyond elastoplastic is an extrapolation, not a tested claim of this paper:

- **Viscoelastic / elastic media** (gels, dough at rest, soft tissue, certain elastomers): probe-induced perturbations are expected to relax; PROBE-THEN-ACT-style probing is plausibly net-positive. Only the elastoplastic case is tested here.
- **Granular media without cohesion** (dry sand, gravel, beads, rice): probe-induced perturbations do not relax. Probing was net-negative on every granular split we tested.
- **Cohesive compacting media** (wet sand, snow): cohesive in appearance but capable of persistent compaction under contact loading. Snow showed a 13.3pp loss in our setup; the broader regime is not tested.

This scope claim suggests a practical stack for mixed materials: use a coarse material-class router that selects between PROBE-THEN-ACT and a reactive policy.

B. Why the Privileged-Parameter Baseline Fails on the Elastoplastic Split

A residual PPO policy maps observation to joint residual, trained over rollouts. M8's training distribution is dominated by sand rollouts where rebound is negligible, so its residual head was never shaped to anticipate elastic rebound. At test time on elastoplastic, reading (E, ν) does not unlock a response curve that was never trained; the sand-shaped residual over-corrects and the rollout fails. PROBE-THEN-ACT sidesteps this by using the probe to *elicit* a rebound response at test time that the encoder summarizes for the policy. We frame the lesson narrowly: in our matched, sand-only-trained setting, raw parameter access did not produce a useful elastoplastic policy whereas the probe trace did. A more carefully designed teacher—trained on a domain-randomized material mix, or given learned features of (E, ν, ρ) rather than raw values—could in principle close this gap.

C. What Does the Encoder Represent?

Because the encoder is trained end-to-end with the policy and never supervised against material labels, z need not resemble a physically interpretable quantity, and we avoid calling it a material embedding. Removing z loses 14.4pp of transfer on elastoplastic, so the latent carries task-relevant signal; characterizing its semantic content (e.g., regression of z against held-out material parameters, or calibration of σ_z) and reusing the encoder across tasks sharing material families are natural next steps.

D. Limitations

The positive result rests on a single OOD split, with no test on other viscoelastic materials (gel, dough, foam) or on training distributions that include any non-sand material. With three seeds and $\sim$40pp per-seed variance on elastoplastic, the 14.7pp gap is directional rather than statistically conclusive. Probing physically degrades granular substrates (the 22pp loss on ID sand is predicted by the hypothesis but is also a hard scope limit: do not deploy PROBE-THEN-ACT on dry granular media). The probe is scripted and identical across materials; a learned probe policy that optimizes an information-gain surrogate may tighten the elastoplastic gain. All results are in Genesis MPM; sim-to-real transfer of probe-style policies on viscoelastic media—especially robustness to real tactile noise and rebound-timescale shifts—is open. Finally, the observation includes simulator-derived aggregate-particle statistics; a sim-to-real deployment would need vision or tactile estimates of those quantities. We use "tactile" throughout in the loose sense of contact-derived response inferred from joints and aggregate motion, not a simulated tactile-array sensor.

E. Connection to Information-Gathering Control

Conceptually, PROBE-THEN-ACT is a coarse instantiation of a broader principle: partition an episode into an information-gathering segment and an execution segment, and back-propagate task reward through both. Related ideas appear in

active learning, Bayesian experimental design, and information-directed sampling in RL. A full treatment with probe length, shape, timing, and material-gated invocation as first-class decision variables is beyond this paper's scope and a natural follow-up.

VII. Conclusion

We introduced PROBE-THEN-ACT as a deliberately minimal experimental probe for active context acquisition under hidden material dynamics, together with a reproducible cross-material edge-push benchmark in Genesis spanning sand, snow, and elastoplastic media (55pp of scripted-transfer spread). All learned policies are trained on sand only and evaluated zero-shot on the four other splits.

The empirical finding is an asymmetric sign pattern. On elastoplastic (recoverable rebound), PROBE-THEN-ACT delivers a 14.7pp transfer gain, a 14.7pp spill reduction, and a 16.7pp success-rate gain over a matched reactive PPO baseline; matched ablations show that removing the probe falls below the reactive baseline, while dropping only the encoder returns to roughly baseline transfer and remains below full PROBE-THEN-ACT. On ID sand, snow, and two sand parameter shifts, the same architecture underperforms by 13–22pp. We organize this pattern through a *recoverable-deformation hypothesis*: probing appears beneficial when probe-induced perturbations relax on the task timescale and detrimental when they leave persistent disruption. A single-seed privileged-parameter diagnostic that scores 0% on elastoplastic adds a complementary observation: in our sand-only-trained setup, an in-episode probing trace was a more useful conditioning signal than a passive parameter vector.

PROBE-THEN-ACT is therefore best read as a phenomenon report and diagnostic instrument, not as a deployment-ready manipulation stack. The recoverability hypothesis makes concrete, falsifiable predictions: matched recoverable/nonrecoverable materials should flip the probe sign with measured post-probe relaxation rather than material label; increasing probe amplitude or reducing settle time should turn an elastoplastic gain negative past a residual-deformation threshold; and equal-capacity dummy probes or non-sand training sources should not create a probe advantage on persistent-perturbation regimes. Failure of these directional tests would reject recoverability as the organizing explanation. Code, configurations, and seeds will be released upon acceptance.

References

[1] Z. Huang, Y. Hu, T. Du, S. Zhou, H. Su, J. B. Tenenbaum, and C. Gan, "PlasticineLab: A soft-body manipulation benchmark with differentiable physics," in *International Conference on Learning Representations (ICLR)*, 2021.

[2] X. Lin, Z. Huang, Y. Li, J. B. Tenenbaum, D. Held, and C. Gan, "DiffSkill: Skill abstraction from differentiable physics for deformable object manipulations with tools," in *International Conference on Learning Representations (ICLR)*, 2022.

[3] H. Shi, H. Xu, S. Clarke, Y. Li, and J. Wu, "RoboCook: Long-horizon elasto-plastic object manipulation with diverse tools," in *Conference on Robot Learning (CoRL)*, 2023.

[4] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, "Domain randomization for transferring deep neural networks from simulation to the real world," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2017.

[5] X. B. Peng, M. Andrychowicz, W. Zaremba, and P. Abbeel, "Sim-to-real transfer of robotic control with dynamics randomization," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2018.

[6] K. Hausman, J. T. Springenberg, Z. Wang, N. Heess, and M. Riedmiller, "Learning an embedding space for transferable robot skills," in *International Conference on Learning Representations (ICLR)*, 2018.

[7] A. Xie, A. Singh, S. Levine, and C. Finn, "Few-shot goal inference for visuomotor learning and planning," in *Conference on Robot Learning (CoRL)*, 2018.

[8] R. Calandra, A. Owens, D. Jayaraman, J. Lin, W. Yuan, J. Malik, E. H. Adelson, and S. Levine, "More than a feeling: Learning to grasp and regrasp using vision and touch," in *IEEE Robotics and Automation Letters (RA-L)*, 2018.

[9] Genesis Authors, "Genesis: A universal and generative physics engine for robotics and beyond," Open-source software, 2024.

[10] R. Bajcsy, "Active perception," *Proceedings of the IEEE*, vol. 76, no. 8, pp. 966–1005, 1988.

[11] S. J. Lederman and R. L. Klatzky, "Hand movements: A window into haptic object recognition," *Cognitive Psychology*, vol. 19, no. 3, pp. 342–368, 1987.

[12] D. Xu, G. E. Loeb, and J. A. Fishel, "Tactile identification of objects using Bayesian exploration," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2013.

[13] M. Bauza, A. Bronars, and A. Rodriguez, "Tactile object pose estimation from the first touch with geometric contact rendering," *Conference on Robot Learning (CoRL)*, 2020.

[14] K.-T. Yu, M. Bauza, N. Fazeli, and A. Rodriguez, "More than a million ways to be pushed: A high-fidelity experimental dataset of planar pushing," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2016.

[15] M. Bauza and A. Rodriguez, "A data-efficient approach to precise and controlled pushing," in *Conference on Robot Learning (CoRL)*, 2018.

[16] A. Dutta, E. Burdet, and M. Kaboli, "Predictive visuo-tactile interactive perception framework for object properties inference," *IEEE Transactions on Robotics*, vol. 41, pp. 1386–1403, 2025.

[17] B. Ai, S. Tian, H. Shi, Y. Wang, C. Tan, Y. Li, and J. Wu, "RoboPack: Learning tactile-informed dynamics models for dense packing," in *Robotics: Science and Systems (RSS)*, 2024.

[18] H. Shi, H. Xu, Z. Huang, Y. Li, and J. Wu, "RoboCraft: Learning to see, simulate, and shape elasto-plastic objects with graph networks," in *Robotics: Science and Systems (RSS)*, 2022.

[19] K. Zhang, B. Li, K. Hauser, and Y. Li, "AdaptiGraph: Material-adaptive graph-based neural dynamics for robotic manipulation," in *Robotics: Science and Systems (RSS)*, 2024.

[20] X. Yang, Z. Ji, and Y.-K. Lai, "Differentiable physics-based system identification for robotic manipulation of elastoplastic materials," *The International Journal of Robotics Research*, vol. 44, no. 13, pp. 2126–2155, 2025.

[21] D. Millard, D. Pastor, J. Bowkett, P. Backes, and G. Sukhatme, "GranularGym: High performance simulation for robotic tasks with granular materials," in *Robotics: Science and Systems (RSS)*, 2023.

[22] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *International Conference on Machine Learning (ICML)*, 2017.

[23] A. Nagabandi, I. Clavera, S. Liu, R. S. Fearing, P. Abbeel, S. Levine, and C. Finn, "Learning to adapt in dynamic, real-world environments through meta-reinforcement learning," in *International Conference on Learning Representations (ICLR)*, 2019.

[24] K. Rakelly, A. Zhou, D. Quillen, C. Finn, and S. Levine, "Efficient off-policy meta-reinforcement learning via probabilistic context variables," in *International Conference on Machine Learning (ICML)*, 2019.

[25] L. Zintgraf, K. Shiarlis, M. Igl, S. Schulze, Y. Gal, K. Hofmann, and S. Whiteson, "VariBAD: A very good method for Bayes-adaptive deep RL via meta-learning," in *International Conference on Learning Representations (ICLR)*, 2020.

[26] A. Nagabandi, G. Kahn, R. S. Fearing, and S. Levine, "Neural network dynamics for model-based deep reinforcement learning with model-free fine-tuning," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2018.

[27] I. Clavera, J. Rothfuss, J. Schulman, Y. Fujita, T. Asfour, and P. Abbeel, "Model-based reinforcement learning via meta-policy optimization," in *Conference on Robot Learning (CoRL)*, 2018.

[28] W. Yu, J. Tan, C. K. Liu, and G. Turk, "Preparing for the unknown: Learning a universal policy with online system identification," in *Robotics: Science and Systems (RSS)*, 2017.

[29] M. Memmel, A. Wagenmaker, C. Zhu, D. Fox, and A. Gupta, "ASID: Active exploration for system identification in robotic manipulation," in *International Conference on Learning Representations (ICLR)*, 2024, oral. [Online]. Available: https://openreview.net/forum?id=jNR6s6OSBT

[30] V. Makoviychuk, L. Wawrzyniak, Y. Guo, M. Lu, K. Storey, M. Macklin, D. Hoeller, N. Rudin, A. Allshire, A. Handa, and G. State, "Isaac gym: High performance GPU-based physics simulation for robot learning," in *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2021.

[31] C. D. Freeman, E. Frey, A. Raichuk, S. Girgin, I. Mordatch, and O. Bachem, "Brax: A differentiable physics engine for large scale rigid body simulation," arXiv preprint arXiv:2106.13281, 2021.

[32] X. Lin, Y. Wang, J. Olkin, and D. Held, "SoftGym: Benchmarking deep reinforcement learning for deformable object manipulation," in *Conference on Robot Learning (CoRL)*, 2021.

[33] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.

[34] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, "Stable-Baselines3: Reliable reinforcement learning implementations," *Journal of Machine Learning Research*, vol. 22, no. 268, pp. 1–8, 2021.